

Problem Statement 03: Illegal Fishing Detection

Background

Illegal, Unreported, and Unregulated (IUU) fishing fleets exploit vast sovereign Exclusive Economic Zones. These industrial operators often evade detection by turning off AIS transponders, spoofing kinematic vessel profiles, and transferring catches to refrigerated cargo reefers in adjacent international waters.

Problem

Sensor data across the EEZ is asynchronous, noisy, and heterogeneous: AIS feeds contain deliberate dropouts, satellite SAR passes occur at discrete intervals under cloud cover, and coastal radar is constrained by line-of-sight. Enforcement agencies must accurately differentiate illicit industrial actors from legitimate domestic fishers while operating under strict patrol fuel budgets.

Aim

Develop a multi-sensor data fusion model that fuses AIS, satellite SAR observations, and coastal radar to detect potential illegal fishing operations using available sensor data.

The model should classify detected objects based on whether they are engaged in fishing activities.

Output

1. Complete, reproducible training and inference source code (train.py, predict.py)
2. Exported model weights/checkpoints (.pt or .onnx)
3. Predictions CSV mapping vessel detections to classification and length estimates

Data

File Structure:

PS3_public/

├── train/

| ├── scene_001/ | ├── VH_dB.tif | ├── VV_dB.

```

| | |— bathymetry.tif
| | |— owiWindSpeed.tif
| | └— ...
| └— ...
|— test/
| |— scene_555/
| | |— VH_dB.tif
| | └— ...
| └— ...
|— train_labels.csv
└— sample_submission.csv

```

Data Description:

Scene Structure (per scene folder):

File	Description	Resolution
VH_dB.tif	VH-polarization SAR image	10m pixels
VV_dB.tif	VV-polarization SAR image	10m pixels
bathymetry.tif	Water depth (meters)	500m pixels
owiWindSpeed.tif	Wind speed (m/s)	500m pixels
owiWindDirection.tif	Wind direction (degrees)	500m pixels
owiMask.tif	Land/ice mask	500m pixels

train_labels.csv:

Column	Description	Type
detect_id	Unique detection ID	int
scene_id	Scene identifier	string
detect_scene_row	Pixel row coordinate	int
detect_scene_column	Pixel column coordinate	int
is_vessel	Is it a vessel? (True/False)	bool
is_fishing	Is it fishing? (True/False)	bool
vessel_length_m	Vessel length (meters)	float
distance_from_shore_km	Distance to shoreline	float

Note: Labels are only in train_labels.csv

Submission Format

File Name: **teamname_PS3.csv**

Required Columns

The submission file must contain exactly 4 columns:

1. **detect_id**
2. **scene_id**
3. **is_fishing**

Example Submission

Detect_id, ,scene_id,is_fishing

5678,scene_555,True

6789,scene_555,False

7890,scene_556,True

Requirements

1. The submission file must contain exactly 3 columns: `detect_id`, `scene_id`, and `is_fishing`.
2. `is_fishing` must be a boolean value (`True/False`).
3. The submission must contain one row per detected object, with no missing values (NaN), and must be saved as `teamname_PS3.csv`.

Evaluation Metrics

Your submission will be scored on:

1. F1-Score: Higher is better (70% weight).
2. ROC-AUC: Higher is better (30% weight).

Final Score = $(0.7 \times \text{F1-Score}) + (0.3 \times \text{ROC-AUC})$

Higher final score wins.